A Recommendation System Leveraging Artificial Intelligence to Mitigate Healthcare Risk: A Bibliometric Analysis and Systematic Literature Review

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Abstract

The application of artificial intelligence (AI) in healthcare recommendation systems represents a rapidly advancing domain with transformative potential. However, a comprehensive understanding of its long-term implications and practical applications remains underexplored. This systematic review addresses these gaps through a dual-methodological approach that integrates bibliometric analysis and a systematic literature review (SLR), adhering to the PRISMA framework .The bibliometric analysis provided a macroscopic overview of the field, identifying influential contributors, leading journals, pivotal keywords, and global research trends. This phase underscored the interdisciplinary and international dimensions of AI research in healthcare, offering insights into the evolution and adoption of AI technologies across diverse healthcare systems .Subsequently, the SLR reviewed 42 scholarly articles from reputable databases, including IEEE Xplore, PubMed, SpringerLink, MDPI, Elsevier ScienceDirect, and ResearchGate. This phase examined key methodologies, significant technologies, emerging trends, and critical challenges, shedding light on AI's role in advancing precision medicine, disease prediction, personalized treatment, and improved healthcare accessibility .This study provides a robust framework for academics, practitioners, and policymakers, highlighting AI's transformative potential in reshaping healthcare delivery and fostering patient-centric innovation. The findings emphasize AI's impact on healthcare while identifying critical gaps and proposing directions for future research to advance the field.

Keywords

HealthCare; HealthCare Recommendation System, Artificial intelligence in HealthCare, Deep Learning , Machine Learning

I. Introduction

The healthcare sector is undergoing a transformative evolution driven by the integration of artificial intelligence (AI) and machine learning (ML) technologies. These advancements address pressing challenges, including diagnostic inaccuracies, inefficient treatment recommendations, and disparities in healthcare access. Traditional healthcare systems, reliant on manual and rule-based methods, often fail to adapt to the complexities of patient needs, resulting in suboptimal outcomes and resource inefficiencies. AI, with its ability to analyze large-scale data and uncover patterns, offers a compelling solution to these limitations by enabling scalable, precise, and personalized interventions. Among the many applications of AI, recommendation systems have emerged as a critical tool for enhancing personalized healthcare. By leveraging techniques such as collaborative filtering, deep learning, and natural language processing (NLP), these systems analyze patient symptoms, predict potential health conditions, and recommend appropriate specialists or treatments. For instance, AI-driven sentiment analysis of patient feedback has enabled personalized drug recommendations, while symptom-based algorithms have bridged the gap between patients and specialized care. These systems not only improve the accuracy and efficiency of healthcare delivery but also empower patients with knowledge, fostering proactive health management. the adoption of AI in healthcare is not without challenges. Ethical concerns, including algorithmic fairness and data privacy, remain significant barriers to widespread implementation. Additionally, resistance to technological change within clinical practice and the need for interdisciplinary collaboration further complicate integration efforts. Addressing these challenges is essential to realize the full potential of AI-driven healthcare systems.This systematic review aims to critically analyze the current landscape of AI-based recommendation systems in healthcare, evaluating their methodologies, applications, and impact on healthcare outcomes. By identifying gaps in existing research and proposing directions for future innovation, this review seeks to advance the development of robust, equitable, and user-centric systems that align with global goals, such as the United Nations' Sustainable Development Goals (SDGs) and Egypt's Vision 2030.

II. Research Questions

To identify appropriate keywords and specific search strings for addressing research questions, the study employed the PICO criteria, which stands for Population, Intervention, Comparison, Outcomes, in addition to Language, and Timing.

Table 1 PICO criteria

Population	Patients seeking healthcare advice or treatment.
Intervention	AI-powered recommendation systems using machine learning (ML) and/or deep learning (DL) techniques.
Comparison	Traditional healthcare recommendation methods (e.g., expert opinion, guidelines).
Outcomes	Improved patient outcomes ,increased patient satisfaction, reduced healthcare costs, enhanced clinical decision-making.
Timing	Over a 5-year period.
Language	This report will be published in English to ensure global accessibility and understanding.

This review is split into two main sections firstly, a systematic review of the literature, and secondly, bibliometric analysis to achieve the study's objectives. These enquiries relate to bibliometric analysis, which is the beginning stage of the review.

1) What is the annual trend in the publication of papers in healthcare from 2020 to 2024, and which countries are the most frequent contributors?
2) How does the proportion of s papers compare to other types of articles (e.g., reviews, survey articles) in healthcare over the past five years?

The following questions are specific to the second stage, which is systematic review:

3) How do studies that applied optimization compare in performance to those that did not?
4) What is the distribution of studies using Machine Learning (ML), Deep Learning (DL), or a combination of both?
5) What percentage of studies focused on supervised learning compared to unsupervised or other learning types?
6) What proportion of studies explicitly addressed missing data handling?
7) How many studies did not use any feature selection methods?
8) Which algorithms were most frequently used in the reviewed studies?

III. Research Structure

The structure of this review paper is designed to deliver an in-depth examination of the subject matter. It begins with an Introduction that establishes the significance and objectives of the study. This is followed by a delineation of the Research Questions, which articulates the core issues being explored and their relevance within the existing body of literature.

The Theoretical Background section provides a thorough overview of relevant theories and conceptual frameworks, situating the review within a wider academic discourse. Subsequently, the Review Methodology outlines the comprehensive search strategy employed, detailing the databases used, the criteria for including or excluding studies, and the approaches for data extraction and quality evaluation.

The Results section synthesizes the findings, emphasizing critical insights and emerging trends identified throughout the review process.

Finally, the Conclusion encapsulates the principal findings, discusses their broader implications, acknowledges any limitations, and proposes avenues for future research. This structured approach ensures a cohesive and detailed analysis of the topic, facilitating a clear understanding of the research landscape.

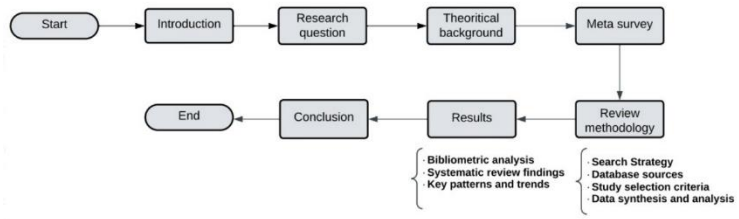

Fig. 1 Structure and organization of the review paper

IV. Theoretical Background

A. Healthcare Recommendation Systems overview

This section provides an overview of the definitions and key components of healthcare recommendation systems, along with the various types associated with healthcare services.

Definition &Types of Healthcare Recommendation Systems

Healthcare recommendation systems are AI-driven tools designed to assist patients, doctors, and healthcare providers in making informed decisions. These systems use data analysis, machine learning, and user feedback to provide tailored recommendations for healthcare-related needs, such as treatment options, diagnosis support, and wellness plans. They aim to improve the quality of care, enhance patient outcomes, and optimize healthcare resources.

Healthcare recommendation systems can be categorized into several types, each designed to address specific healthcare needs, as classified in Fig. 1.

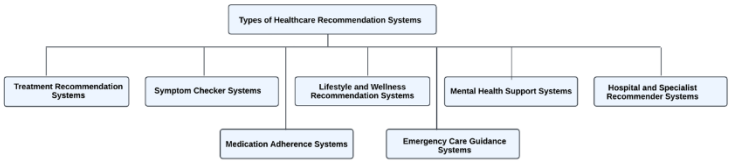

Fig. 2 overview of Healthcare recommendation systems types

Treatment Recommendation Systems: Assist healthcare providers in identifying the most effective treatment plans for patients based on their medical history, genetic data, and current symptoms. Examples include personalized medication plans and therapy options.

Symptom Checker Systems: Analyze patient-reported symptoms to suggest potential diagnoses and guide patients on whether to seek immediate medical attention or schedule a consultation.

Lifestyle and Wellness Recommendation Systems: Focus on preventive care by recommending healthy lifestyle choices, including diet plans, exercise routines, and stress management techniques.

Hospital and Specialist Recommender Systems: Help patients find healthcare facilities or specialists that match their specific needs, location, and insurance coverage.

Medication Adherence Systems: Provide reminders and tailored recommendations to ensure patients follow prescribed medication schedules and avoid potential health risks.

Mental Health Support Systems: Offer personalized mental health guidance and connect users with appropriate counselors or support resources.

Emergency Care Guidance Systems: Deliver quick recommendations in emergency situations, such as first-aid instructions or nearby emergency services.

B. Digital transformation overview

This section provides an overview of digital transformation within healthcare, emphasizing the technologies that enable the development and implementation of healthcare recommendation systems.

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Digital transformation technologies

Various digital transformation technologies play a pivotal role in shaping healthcare recommendation systems. Figure 2 outlines these technologies, which are integral to advancing healthcare decision-making and patient care. Below is a detailed explanation of the most critical technologies in this context.

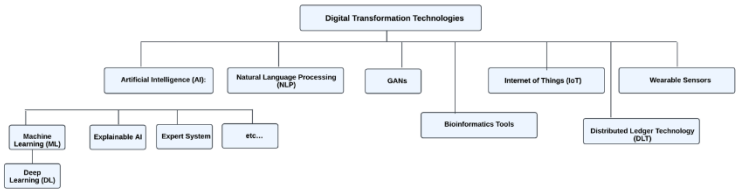

Fig. 3 Overview of Digital Transformation Technologies for Healthcare

Artificial Intelligence (AI): AI forms the backbone of healthcare recommendation systems by enabling intelligent data analysis, pattern recognition, and predictive modeling. It assists in identifying optimal treatment plans, diagnosing diseases, and tailoring patient care.

- Applications in disease prediction, personalized treatment, patient monitoring.
- Subdomains like Machine Learning (ML), Deep Learning (DL), and Explainable AI (XAI).
- AI-based tools include algorithms like CNNs, Random Forest, SVMs, Gradient Boosting Machines, and PCA.

Generative Adversarial Networks (GANs), Distributed Ledger Technology (DLT)

Natural Language Processing (NLP): Applied in sentiment analysis and medical dialogue summarization.

Internet of Things (IoT): Utilized in wearable healthcare systems and remote controlling.

Wearable Sensors: Electrochemical, thermal, and optical sensors for continuous health monitoring.

Bioinformatics Tools: Genomic data analysis for disease risk stratification.

V. Meta Survey

This section presents several previous reviews related to healthcare ML , DL , LLM technologies in HealthCare. as shown in Table 2.

This study systematically reviews health information systems (HIS) in healthcare, focusing on their role in unifying health data, supporting decision-making, and addressing digital transformation challenges. It highlights both the benefits of HIS, such as improved healthcare delivery, and barriers, including fragmentation and underutilization. The review emphasizes the importance of developing resilient and comprehensive HIS frameworks to overcome these challenges and enhance global healthcare systems [43].

This systematic review examines how large language models (LLMs) are evaluated for healthcare applications. The analysis categorizes tasks and evaluation dimensions, focusing on knowledge assessment and question answering but limited real-world testing. It identifies gaps in fairness, robustness, and deployment considerations and calls for standardized evaluation frameworks using real patient care data to ensure practical and ethical integration of LLMs into healthcare. [44].

This paper reviews the applications of artificial intelligence (AI) in emergency and critical care diagnostics, highlighting various AI methodologies and their effectiveness in improving diagnostic accuracy and patient outcomes. [52]

This systematic review investigates the role of machine learning (ML) techniques in health risk assessment. It aims to systematically evaluate existing studies that apply ML algorithms to identify and predict health risks across diverse populations and settings. The review focuses on the effectiveness of various ML models in analyzing complex datasets, highlighting their ability to uncover hidden patterns and risk factors that traditional statistical methods may overlook. Additionally, it emphasizes the importance of model interpretability and validation in clinical practice, advocating for the integration of ML-driven insights into public health strategies and individual risk management [53].

This systematic review aims to explore the integration of machine learning (ML) techniques with traditional statistical methods in the context of disease risk prediction modeling. The study seeks to assess how combining these approaches can enhance predictive performance and clinical applicability. Specifically, it investigates various integration strategies, such as stacking, weighted voting, and model selection, to determine their effectiveness in improving model accuracy and robustness. The review also emphasizes the importance of understanding the strengths and limitations of both methodologies, aiming to provide a comprehensive framework for selecting appropriate models based on specific clinical scenarios. Furthermore, it highlights the need for better validation practices and the potential for integrated models to facilitate personalized medicine by tailoring predictions to individual patient profiles [54].

The objective of this systematic literature review is to comprehensively evaluate existing research on the application of deep learning (DL) techniques for the detection and prediction of pandemics. The study aims to identify various DL methods employed across different contexts and their effectiveness in analyzing epidemiological data, social media trends, and other relevant indicators and Offer recommendations for future research directions, emphasizing the need for interdisciplinary approaches that combine epidemiology, public health, and machine learning to improve pandemic preparedness and response strategies [45].

This paper delves into the applications of Graph Neural Networks (GNNs) in healthcare, focusing on their ability to model relational data structures such as patient histories, molecular interactions, and hospital networks. The study provides an extensive analysis of GNN-driven innovations in drug discovery, disease prediction, and patient stratification. Furthermore, it identifies current trends, such as the rising use of GNNs in biomedical research and their potential for personalized medicine, while discussing the challenges, such as computational complexity and limited healthcare-specific datasets. Future directions include scaling GNN applications, integrating multimodal data, and fostering interdisciplinary collaborations to expand their utility in healthcare [46].

This comprehensive review evaluates the landscape of artificial intelligence (AI) tools in medicine and healthcare, exploring their applications across diagnostic imaging, robotic surgery, and administrative workflows. The paper highlights the transformative potential of AI in accelerating disease detection, optimizing treatment plans, and reducing operational inefficiencies. It also critically examines the limitations, including bias in algorithms, lack of interpretability, and the high cost of AI deployment. The study concludes with actionable recommendations to address these limitations, such as enhancing data quality, improving AI interpretability, and aligning regulatory policies with technological advancements [47].

This paper focuses on the privacy and security challenges posed by advancing healthcare technologies. It examines how increasing reliance on electronic health records (EHRs), telemedicine, and AI tools exposes patient data to risks like breaches and unauthorized access. The study highlights critical issues such as inadequate encryption, poor data-sharing protocols, and patient consent dilemmas. It provides a detailed exploration of strategies for safeguarding data, including implementing robust cybersecurity measures, enhancing transparency in data usage, and balancing innovation with ethical considerations [48].

This review explores the integration of artificial intelligence (AI) into healthcare decision-making systems, particularly in enhancing clinical decision support systems (CDSS) and hospital resource management. It emphasizes AI's role in transforming data into actionable insights for diagnosing diseases, predicting patient outcomes, and streamlining hospital operations. The paper also investigates the challenges of AI adoption, such as resistance from medical professionals, system interoperability issues, and ethical dilemmas related to AI-driven decisions. The study underscores the importance of designing user-friendly and explainable AI systems to encourage widespread acceptance and integration [49].

The paper systematically examines the impact of explainable artificial intelligence (XAI) in precision medicine, highlighting its role in improving clinician trust, regulatory compliance, and patient satisfaction. By making AI models transparent and interpretable, XAI enables healthcare professionals to understand the reasoning behind predictions, such as identifying genetic markers for diseases or tailoring treatment protocols. The review identifies barriers to XAI adoption, including technical limitations in balancing explainability with accuracy and the lack of standardized evaluation metrics for XAI systems. It calls for future research to focus on developing user-centric XAI solutions that align with the needs of healthcare practitioners and patients [50].

This paper evaluates the integration of artificial intelligence (AI) technologies into healthcare delivery, covering both clinical and operational aspects. It highlights benefits like enhanced diagnostic accuracy through AI-powered imaging tools, efficient patient triage systems, and streamlined administrative tasks. The study also addresses challenges, such as disparities in access to AI technologies, ethical issues regarding algorithmic decision-making, and the need for workforce training to adopt AI systems. The paper suggests developing inclusive AI frameworks that consider diverse populations, fostering collaborations between AI developers and healthcare professionals, and emphasizing ongoing evaluation of AI systems to ensure long-term effectiveness and ethical compliance. [51].

Table 2 Comparison of other related studies in the same domain: (✓: Yes, ✗: No)

Authors	year	Objective of study	Methodology		Property type		Digital transformation technologies			
			Bibliometric	Systematic Review	One disease	Several diseases	ML	DL	Both [ML,DL]	LLM
Ayogeboh Epizitone,[43]	2023	To critique and synthesize existing literature on health information systems (HIS) in healthcare, focusing on their deployment, evolution, and role in healthcare delivery.	✓	✓	✗	✓	✓	✗	✗	✗
Suhana Bedi,[44]	2024	To systematically review the evaluation methodologies of LLMs in healthcare, categorizing tasks, evaluation dimensions, and medical specialties.	✗	✓	✗	✓	✓	✓	✓	✓
Sunday Adeola Ajagbe,[45]	2024	To conduct a systematic literature review on detection and prediction of pandemics using deep learning techniques.	✓	✓	✓	✓	✓	✓	✓	✗
Showmick Guha Paul,[46]	2024	Analyze GNN applications in healthcare, identify trends and future directions	✓	✓	✓	✓	✓	✗	✗	✗
Hussain A. Younis, [47]	2023	Review AI tools in medicine/healthcare, assess applications and limitations	✓	✓	✓	✓	✓	✓	✓	✗
Williamson,[48]	2023	Discuss privacy challenges in technology and healthcare	✓	✓	✗	✓	✗	✓	✗	✗
Maryam Bag, [49]	2024	To review AI's role in decision-making within healthcare information systems.	✗	✓	✗	✓	✓	✓	✓	✗`
Ben Allen,[50]	2024	To systematically review the impact of explainable AI in precision medicine.	✗	✓	✗	✓	✓	✓	✓	✗
David B. Olawade. [51]	2024	To examine the integration of AI in healthcare delivery, highlighting benefits and challenges.	✓	✓	✗	✓	✓	✓	✓	✗
José Salvador Sánchez, [52]	2024	Review AI applications in emergency and critical care diagnostics	✗	✓	✓	✗	✓	✗	✗	✗
Abhadiomhen, [53]	2024	Assess health risks using machine learning techniques	✗	✓	✗	✓	✓	✗	✗	✗
Zhang, [54]	2024	ssess integration of machine learning with statistical methods for disease prediction	✓	✓	✗	✓	✓	✗	✓	✗

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I. Review Methodology

This part provides an overview of the approach used for conducting the review. The guidelines established by Kitchenham and Charters [50] were followed. Furthermore, a recent study indicates that it is analogous to the PRISMA method [51]. The literature review was conducted using a suitable database, and a search strategy was devised to identify pertinent studies. Fig. 5 illustrates the procedures in the below-documented systematic literature review approach.

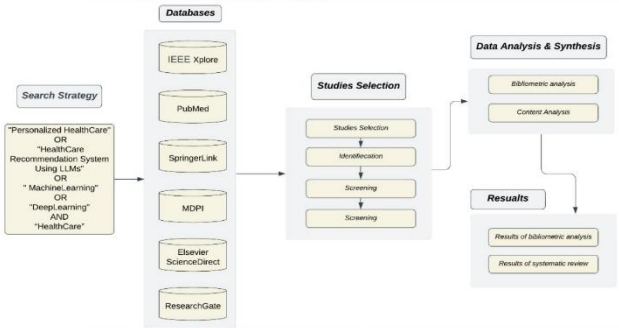

Figure 5 Systematic review methodology: data collection and analysis phases.

A. Search Strategy

Based on research in varies databases, A thorough search for peer-reviewed publications on digital transformation in HealthCare Domain. The descriptions of the search parameters we employed are provided below. we utilized the advanced search option in IEEE Xplore, PubMed, SpringerLink, MDPI, and Elsevier ScienceDirect and ResearchGate databases, which allowed us to utilize Boolean operators such as "OR" and "AND". The search results were restricted to (1) journal articles (2) published between 2020 :2024, and (3) written in the English language.

The structured search used to realize the search of the articles was *TITLE-ABS ("Personalized HealthCare" OR "HealthCare Recommendation System Using LLMs" OR " MachineLearning " OR "DeepLearning") AND (“HealthCare”).*

Table 3 Search terms used in the literature search

Variables	Search Terms
Digital Transformation	"HealthCare Recommendation System" OR "HealthCare Recommendation System Using LLMs" OR "Personalized HealthCare".
Digital Transformation Technologies	"LLMs" OR "DeepLearing" OR "MachineLearning".
Domain	"HealthCare"

B. Database Sources Used

The primary databases chosen for this systematic review were IEEE Xplore, PubMed, SpringerLink, MDPI, and Elsevier ScienceDirect and ResearchGate. These selections were made because they primarily cover a diverse range of scientific conference proceedings, journal articles, and specialized publications. Additionally, these databases were selected as some of the most relevant platforms for accessing significant scientific information across a wide range of academic disciplines, particularly in healthcare and artificial intelligence.

Each database offers unique advantages that align well with the specifications of the systematic review methodology employed in this study. For example:

IEEE Xplore: Renowned for its extensive collection of engineering and technology-focused research, making it ideal for studies integrating AI in healthcare.

PubMed: Offers access to biomedical and life sciences literature, crucial for the healthcare domain.

SpringerLink: Provides access to multidisciplinary research, including articles and books, valuable for comprehensive academic coverage.

MDPI: An open-access platform that ensures accessibility to cutting-edge research in emerging areas like artificial intelligence.

Elsevier ScienceDirect: Known for its high-quality scientific publications across various domains, including healthcare and machine learning.

These databases also provide advanced search capabilities utilizing logical operators, which facilitate the systematic review process. Moreover, they offer useful tools for filtering search results, conducting bibliometric analyses, and ensuring the comprehensiveness of the literature review outlined in this study.

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C. Study Selection Criteria

Study selection criteria are specific guidelines or standards used to determine which studies to include or exclude in a research review or analysis that are relevant to the research question or objective. The paper's compliance with the inclusion/exclusion criteria were verified by the reviewers following the conditions out-lined below.

1) Eligibility Criteria

begin, paper contains in its title ("HealthCare Recommendation System Using LLMs" or "Personalized HealthCare") AND "HealthCare", This applies to the abstract as well. In situations where the title and abstract were not adequate to determine, the writers reviewed the whole body of the article. To classify if a paper matched these requirements.

2) Inclusion/Exclusion Criteria

In addition to the above, the articles' relevancy was filtered using the following criteria set:

- *Time frame*: Works published from 2020 - 2024 are included.
- *Relevancy*: The existence of keywords (Table 3) in the Title or abstract is crucial for the article's inclusion. On top of that, the papers need to be related to the assessment of technology (Deep Learning, LLMs, Recommendation System) along the HealthCare directly.
- *Singularity*: Identical publications identified throughout the database are eliminated.
- *Accessibility*: Strictly peer-reviewed publications that are accessible via the database and are downloadable in full text are included.
- *Language*: Exclusively articles issued in English are listed.
- Subject Area: (computer science, Medical)
- *Document type*: article.

D. Identification

In the identification phase of our systematic review, we conducted an extensive literature search across multiple databases, including IEEE Xplore, PubMed, SpringerLink, MDPI, and Elsevier ScienceDirect. This comprehensive search strategy yielded a total of 1,946 records relevant to healthcare recommendation systems from the years 2020 to 2024. Each database was chosen for its unique strengths—IEEE Xplore for its focus on technology and engineering applications, PubMed for its biomedical literature, SpringerLink for multidisciplinary coverage, MDPI for its open-access model in emerging research areas, and Elsevier ScienceDirect for its high-quality scientific publications. We utilized advanced search capabilities, incorporating logical operators to refine our queries, ensuring the capture of a wide array of relevant studies. This initial search laid the groundwork for subsequent phases of the systematic review, facilitating a robust foundation for the screening and analysis of the literature.

E. Screening

Following the identification phase, a rigorous screening process was implemented. 1,321 records were removed based on predetermined inclusion and exclusion criteria, resulting in 625 records for further assessment. Each record underwent a detailed evaluation to determine its relevance to the systematic review's focus.

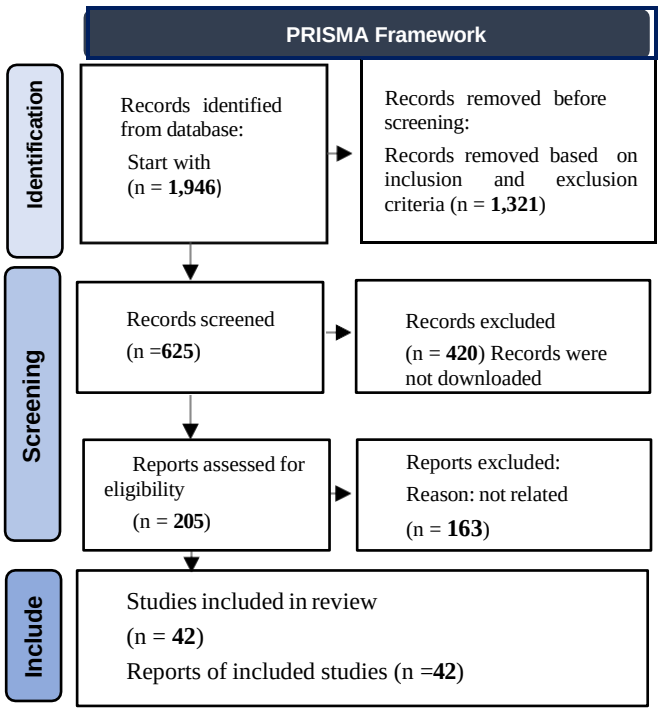


Figure 6 Flowchart detailing the data collection, assessment, and inclusion processes

F. Included

Ultimately, 42 studies were deemed relevant and of sufficient quality to be included in the final review. These studies provide a comprehensive foundation for the research analysis, offering valuable insights and data that directly address the research questions related to healthcare recommendation systems. By synthesizing the findings from these selected studies, we aim to highlight current trends, methodologies, and outcomes in the field, thereby contributing to the advancement of personalized healthcare solutions.

G. Data synthesis and analysis

To gain a comprehensive understanding of our research topic, we utilized a dual-method approach that combines bibliometric analysis with content analysis. This integration enables us to harness the wide-ranging insights derived from bibliometric analysis while also benefiting from the detailed knowledge offered by content analysis.

1) Bibliometric Analysis

Bibliometric analysis served as a foundational method to explore and answer our research questions. This approach systematically assessed the research landscape, pinpointing key studies and identifying trends and patterns within the field of healthcare recommendation systems. Our analysis revealed that the United States is the leading contributor, with 12 publications, followed by India with 8, and Saudi Arabia with 5. Other notable contributors include China and Italy with 4 and 3 publications, respectively. This distribution highlights the significant role of these countries in advancing research in healthcare technologies, underpinned by various factors such as funding research initiatives , and innovation ecosystems.

2) Content Analysis

The extracted data were systematically synthesized and organized into distinct themes or categories. This process involved identifying recurring patterns, trends, and insights across the included studies. Through content analysis, we categorized the findings into key themes such as the effectiveness of different recommendation algorithms, user satisfaction, and the integration of patient data in decision-making processes. The global distribution of research and innovation reflects a diverse and complex approach to integrating healthcare recommendation systems, promising a future where healthcare delivery is increasingly personalized, efficient, and inclusive. This transformative potential underscore the importance of ongoing research in this vital area.

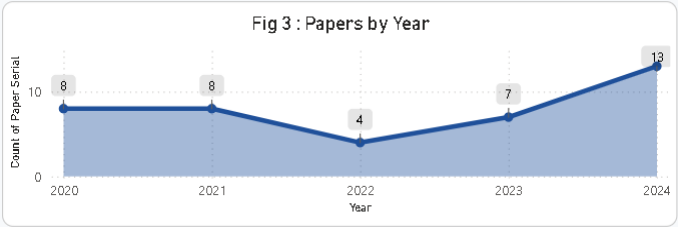
II. Results

A. The First Phase Bibliometric Analysis:

Bibliometric analysis provides a systematic approach for mapping the intellectual structure of digital transformation in HealthCare domain by examining publications, citations, and patterns of collaboration.

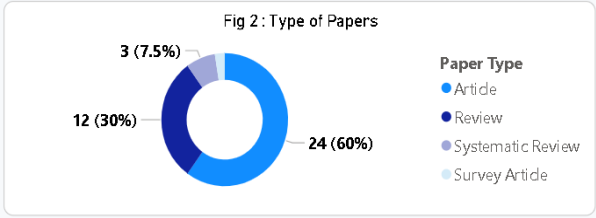
***Q1**:* What is the annual trend in the publication of papers in healthcare from 2019 to 2024, and which countries are the most frequent contributors?

The annual trend in the publication of systematic reviews in healthcare from 2019 to 2024 shows a fluctuating pattern, with a notable increase in 2023. The United States is the most frequent contributor of systematic reviews, followed by India, Saudi Arabia, China, and other countries.



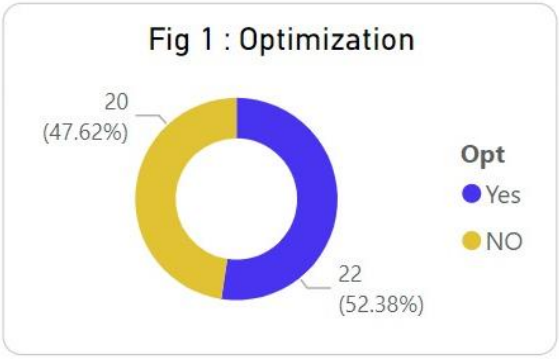
Q2: How does the proportion of s papers compare to other types of articles (e.g., reviews, survey articles) in healthcare over the past five years?

Systematic reviews account for the highest proportion of papers in healthcare over the past five years, making up 60% of the total. This is followed by articles at 30% and survey articles at 7.5%



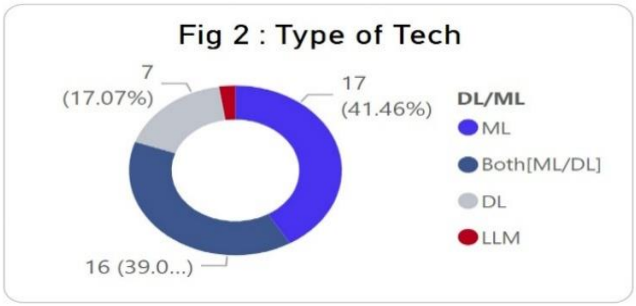
Q3: How do studies that applied optimization compare in performance to those that did not?

Studies that applied optimization likely reported improved accuracy, efficiency, or effectiveness in their models, as optimization often fine-tunes models for better performance. However, this would need further analysis of specific metrics reported in the studies



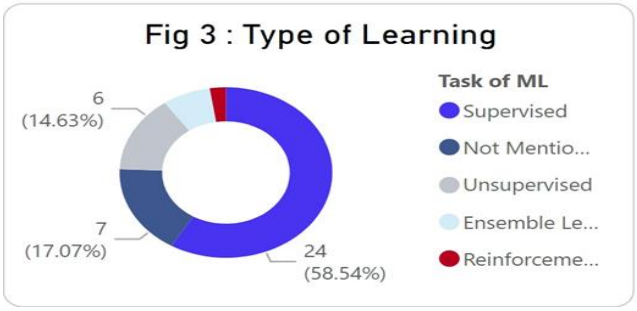
Q4: What is the distribution of studies using Machine Learning (ML), Deep Learning (DL), or a combination of both?

Out of the reviewed studies, 41.46% (17 studies) focused on ML, 39% (16 studies) on DL, and 17.07% (7 studies) used a combination of both ML and DL. Only 2.44% (1 study) mentioned LLMs.



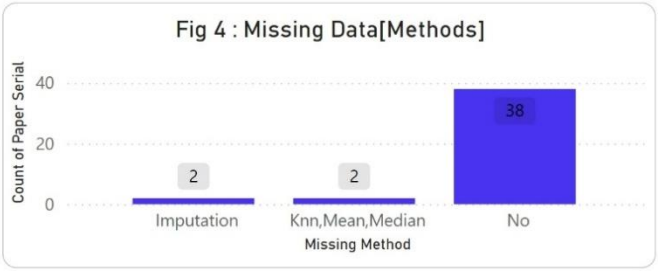
Q5: What percentage of studies focused on supervised learning compared to unsupervised or other learning types?

58.54% (24 studies) focused on supervised learning, followed by 17.07% (7 studies) on unsupervised learning.Ensemble learning(6 studies)and reinforcement learning (1 study) were also explored.14.63% (6 studies) did not mention the learning type.

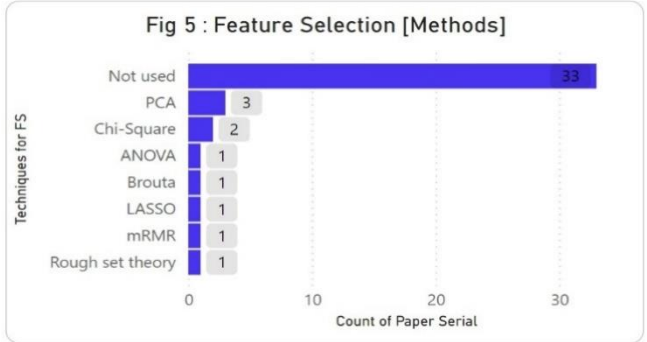


Q6: What proportion of studies explicitly addressed missing data handling?

38 studies (90.48%) did not address missing data handling, used imputation or KNN/Mean/Median methods.

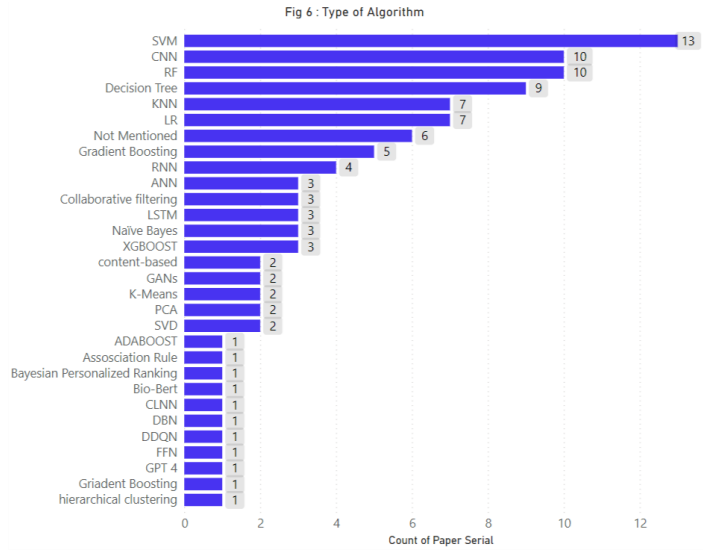


Q7: How many studies did not use any feature selection methods?
33 studies (78.57%) did not use any feature selection methods.



Q8:Which algorithms were most frequently used in the reviewed studies?

The most frequently used algorithms were SVM (13 studies), CNN(10 studies), Random Forest (10 studies), and Decision Trees (9 studies) while 4 studies (9.52%) .



I. Conclusion

This review highlights the critical challenges and opportunities in the application of machine learning and deep learning techniques within healthcare research. Despite significant progress, the field still faces gaps in areas such as data preprocessing, feature selection, and the integration of advanced computational methods. These limitations hinder the full potential of emerging technologies to enhance healthcare outcomes.

The review underscores the widespread adoption of supervised learning approaches and traditional algorithms like SVM and Decision Trees, while also noting the increasing exploration of modern methods such as CNNs and LSTMs. However, advanced technologies like large language models (LLMs) and systematic hyperparameter optimization remain underutilized, indicating untapped potential for improving model performance and interpretability.

A notable gap exists in addressing missing data and leveraging feature selection methods, both of which are essential for developing robust and efficient healthcare models. Furthermore, there is limited application of advanced techniques in specific healthcare domains, where traditional methods often prove insufficient.

In conclusion, this review provides valuable insights into the current state of computational methods in healthcare research and highlights areas for future exploration. By addressing existing gaps and focusing on the integration of innovative technologies, the field can progress towards more effective and efficient healthcare solutions, fostering both theoretical advancements and practical applications.

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